PARVATI VISWANATHAN

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PROFILE

Master's student in Computer Science with research interests in Machine Learning, Agentic AI, Interpretability, and AI safety. Interested in building trustworthy and interpretable intelligent systems.

EDUCATION

UNIVERSITY OF VIRGINIA Charlottesville, VA

MS (Thesis), Computer Science | GPA: 3.87/4.0 Aug 2025 - Present (Expected Dec 2026)

Relevant Coursework: Natural Language Processing, ML for Image Analysis, Neural Networks, Geometry of Data, Probabilistic ML, Convex Optimization

SREE CHITRA THIRUNAL COLLEGE OF ENGINEERING (SCTCE) Thiruvananthapuram, India

BS, Computer Science |Cumulative GPA: 9.42/10.0 October 2020 - Jun 2024

Relevant Coursework: Programming in C, Operating Systems, Analysis of Algorithms, Data Analytics, AI, Mathematics for ML, Deep Learning, Linear Algebra and Calculus, Probability, Graph Theory

- Graduated in top 5 percent of my batch, Class topper in Semesters 1, 2 and 5
- Dr. Ruchi Sharma Endowment Scholarship
- Arun-Aneesh Memorial Award

RESEARCH

Master's Thesis | Charlottesville, VA Jan 2026 – Present

- Working with Prof. Ferdinando Fioretto on a benchmark for testing the endogenous misalignment of self-evolving agents.
- Literature review on existing benchmarks for testing agentic misalignment using long-horizon task sequences.
- Worked on the testing of memory management and update behaviours for a personal assistant agent, probing for evolution-induced safety failures and performing Chain-of-Thought analysis on trajectories failing safety.

Research Assistant, University of Virginia | Charlottesville, VA Aug 2026 – Present

- Probing for misalignment in self-evolving agents

Junior Research Fellow, IISER Thiruvananthapuram | Thiruvananthapuram, India Jun 2024 – Jun 2025

- Worked in the School of Data Science under Dr. Mainak Adhikari on developing an optimized federated learning algorithm using momentum-based techniques and weighted aggregation to handle heterogeneous client data.
- Validated the algorithm on an edge-based data-driven decision system for irrigation using 2 autonomous agricultural rovers.
- Worked on an Actor–Critic model based approach using PyTorch to dynamically control training intensity across clients in each Federated Learning round.

Undergraduate Research, SCTCE | Thiruvananthapuram, India Sep 2023 – Dec 2023

- Conducted a survey of state-of-the-art methodologies of SLAM and various existing applications of SLAM in Augmented Reality. Delivered a seminar to present and discuss key findings from the survey.

PUBLICATIONS & PATENTS

- **SEABench:** Benchmarking Endogenous Misalignment in Self-Evolving Agents **| ICLR 2027 (Under Review)**
- Federated Learning using Nesterov Momentum for Non-IID Data Handling in Edge Networks | **IEEE TCSS 2026**
- An Adaptive Federated Deep Reinforcement Learning with Bayesian Optimization for Data Analysis in Edge Networks | **DACS 2024 (IIM Bodh Gaya)**
- **Indian Patent:** Patent Application 202541052562 **(Under Review):** An Autonomous Rover for Real-Time Analysis of Irrigation Requirement in an Agricultural Field and Method Thereof

SKILLS & COURSES COMPLETED

Skills

- Languages: Python, Java, SQL, C, HTML, CSS, Bash
- Tools: Excel, Visual Studio Code, Git

- Frameworks/Libraries: PyTorch, TensorFlow, HuggingFace, OpenAI, LangChain, ARCore, Numpy, Scikit-learn, Pandas
- Platforms : Linux, Windows, Arduino, Raspberry Pi, Docker

Courses completed
- Coursera: Mathematics for Machine Learning, Programming in Python, Python Data Structures
- NPTEL: Computer Vision, Deep Learning

WORK EXPERIENCE

Teaching Assistant, University of Virginia | Charlottesville, VA May 2026 – Present
- DS 5110 (Big Data Systems): Responsibilities include grading, holding office hours, holding lab sessions
- APMA 3100 (Probability): Responsibilities are mainly grading for a group of 120 students along with other TAs

Teaching Assistant, IISER Thiruvananthapuram | Thiruvananthapuram, India Jan 2025 – Jun 2025
- Database Management Systems: Responsibilities included guiding SQL-based labs, grading assignments, and preparing quizzes for a class of 60 students

Transportation and Business Unit Intern | Thiruvananthapuram, India Oct 2023 – Jun 2024
- Training in Automotive Electronics & Operating systems, Microcontrollers

TECHNICAL EXPERIENCE & PROJECTS

Bachelor's Projects

Campus Navigation System using SLAM and Augmented Reality, SCTCE Sept 2023 - May 2024
- Implemented an AR markerless campus navigation system using SLAM toolkits (ARCore) in Java, and worked on real-time AR localization and visualization.

Chatbot for Legal Queries, SCTCE Jan 2024 - Apr 2024
- Developed a Python-based chatbot using Llama and RAG, using the Constitution of India and documents of other legal cases as the document corpus.

LEADERSHIP & VOLUNTEERING

Women in Power coordinator, Executive Committee, IEEE PES SCT Student Branch Feb 2023 - Feb 2024
- Took the lead in organizing a school outreach program for 40 girls from a government school in Thiruvananthapuram as part of Women in STEM initiatives
- Was part of a 4-person core committee for organizing an inter-college technical fest exclusively for girls, which saw an overall registration of 300+ participants across the state

Documentation Manager, Student Team, IEEE Computer Society Kerala Chapter Feb 2023 - Feb 2024
- Actively managed chapter documents, reports, proposals, and records of chapter activities

National Service Scheme Volunteer, NSS SCTCE (2 yrs)
- Actively involved in community outreach, cleaning drives, and student networking programs, culminating in a 7 day camp focused on community service

Kerala Flood Relief Volunteer
- State Government Disaster Management Volunteer Program, July 2020